INCH-POUND
MIL-DTL-24705/3A(SH)
23 August 2010
SUPERSEDING
MIL-P-24705/3(SH)
7 June 1990

DETAIL SPECIFICATION SHEET

PENETRATORS, MULTIPLE CABLE, ELECTRIC CABLE, ROUND FRAME, SQUARE-FACED BLOCKS

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification sheet and MIL-DTL-24705.

ROUND MULTIPLE CABLE PENETRATORS.

Round multiple cable penetrator assembly (figure 1). Components required for each complete round metal frame multiple cable penetrator assembly specified by this specification sheet shall be:

a. One round frame (see figure 2).
b. The appropriate round end-packing assembly, compression plates, and all hardware (see figure 3).
c. Square-faced cable insert blocks and blanking insert blocks to fill the interior of the round end-packing assembly (see MIL-DTL-24705/1).

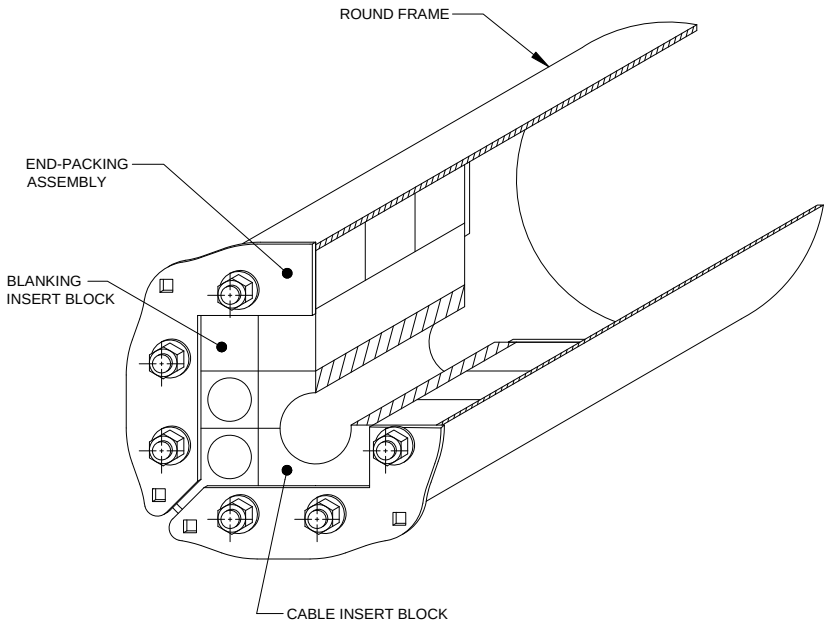


FIGURE 1. Round multiple cable penetrator, assembled view – typical.

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ASSOCIATED COMPONENTS AND THEIR PART OR IDENTIFYING NUMBERS (PINS).

NOTES:

1. Dimensions are in millimeters.
2. Inch equivalents are based on 25.4 millimeters equal to one inch and all are given in parenthesis for general information only.
3. Unless otherwise specified herein, tolerances are ±0.5 mm (±0.02).
4. Unless otherwise specified herein, tolerances for molded rubber parts shall be in accordance with the commercial tolerance (A3) specified in MIL-HDBK-149, Standard Dimensional Tolerances - Molded Solid Rubber Products.
5. All threads shall be in English dimensions. No metric threads shall be supplied.

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FRAMES. Round multiple cable penetrator frames shall be in accordance with figure 2 and the dimensions specified in table I.

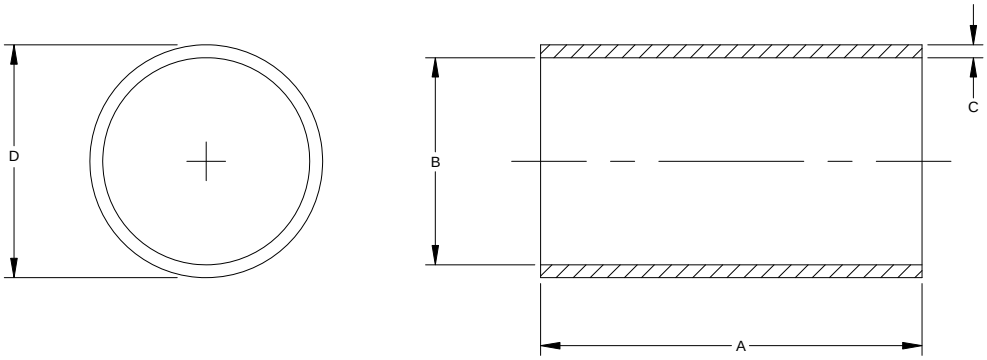


FIGURE 2. Round frame.

TABLE I. Round frame dimensions.

Designator	A (Deck)	A (Bulkhead)	B	C	D
F*70	305 (12.0)	102 (4.0)	74 (2.91)	See dimensions for round frame below	See dimensions for round frame below
F*75	305 (12.0)	102 (4.0)	79 (3.11)	See dimensions for round frame below	See dimensions for round frame below
F*100	305 (12.0)	102 (4.0)	103 (4.05)	See dimensions for round frame below	See dimensions for round frame below
F*125	305 (12.0)	102 (4.0)	129 (5.7)	See dimensions for round frame below	See dimensions for round frame below
F*127	305 (12.0)	102 (4.0)	131 (5.15)	See dimensions for round frame below	See dimensions for round frame below
F*150	305 (12.0)	102 (4.0)	154 (6.06)	See dimensions for round frame below	See dimensions for round frame below
F*200	305 (12.0)	102 (4.0)	203 (7.99)	See dimensions for round frame below	See dimensions for round frame below
NOTE: The asterisk (*) in the designator in table I represents the item material.					

Part or identifying number (PIN) for round frame. Frames are differentiated by an item designation consisting of one to three digits corresponding to a particular size. An example of a round frame PIN is provided below (see MIL-DTL-24705 for complete details).

M	**24705**	**/3**	**-**	**F**	**S**	**75**
Prefix for Military Specification	Specification Number	Spec Sheet No.		Item Type	Item Material	Item Designation

Materials, round frame. The material shall be CRES (C) or steel (S) in accordance with MIL-DTL-24705.

Dimensions, round frame. Round frames shall be fabricated from pipe equivalent to American National Standard schedule 40 or schedule 80 welded and seamless wrought pipe. Dimensions “C” (wall thickness) and “D” (outside diameter) shall be equivalent to the dimensions for American National Standard schedule 40 or schedule 80 welded and seamless wrought pipe selected for the required application.

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END-PACKING ASSEMBLY. Round end-packing assemblies shall be in accordance with figure 3 and the dimensions specified in table II.

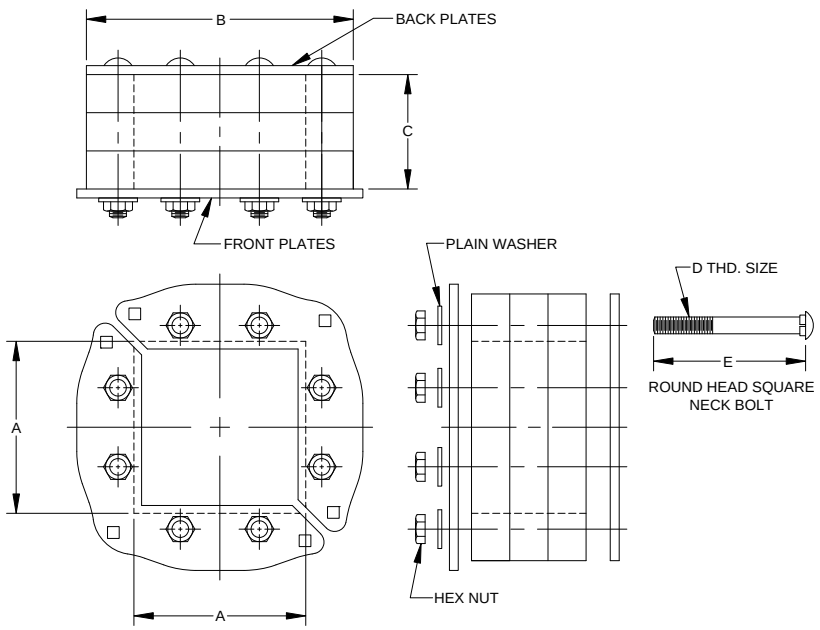


FIGURE 3. Round end-packing assembly (typical).

TABLE II. Round end-packing assembly dimensions.

Designator	A	B	C	D	E
J*70	40.0 (1.57)	70.0 (2.75)	66.0 (2.60)	0.250-20 UNC	85.7 (3.38)
J*75	40.0 (1.57)	75.0 (2.95)	66.0 (2.60)	0.250-20 UNC	85.7 (3.38)
J*100	60.0 (2.36)	100.0 (3.93)	66.0 (2.60)	0.250-20 UNC	85.7 (3.38)
J*125	80.0 (3.15)	125 (4.92)	69.0 (2.72)	0.312-18 UNC	88.9 (3.50)
J*127	80.0 (3.15)	127 (5.0)	69.0 (2.72)	0.312-18 UNC	88.9 (3.50)
J*150	90.0 (3.54)	150.0 (5.90)	69.0 (2.72)	0.312-18 UNC	88.9 (3.50)
J*200	120.0 (4.72)	200.0 (7.87)	69.0 (2.72)	0.312-18 UNC	88.9 (3.50)

NOTE: The J in the designator in table II represents the types of round end-packing assembly specified – E, H, and K (see MIL-DTL-24705). The asterisk (*) in the designator in table II represents the item material which pertains to hardware only.

Part or identifying number (PIN) for round end-packing assembly. End-packing assemblies are differentiated by an item designation consisting of one to three digits corresponding to a particular style. An example of a round end-packing assembly PIN for a square block round frame packing assembly with steel hardware is provided below (see MIL-DTL-24705 for complete details).

M	**24705**	**/3**	**-**	**E**	**S**	**100**
Prefix for Military Specification	Specification Number	Spec Sheet No.		Item Type	Item Material	Item Designation

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Material, round end-packing assembly. The item hardware material shall be CRES (C) or steel (S) in accordance with MIL-DTL-24705.

Design, round end-packing assembly. Item designators J*70 through J*100 shall have four bolts, nuts, and washers in the assembly and item designators J*125 through J*200 shall have eight bolts, nuts, and washers in the assembly.

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REQUIREMENTS.

Design performance.

a.	Nonmagnetic	When specified (see MIL-DTL-24705)
b.	Durometer	81±6
c.	Compression	2 percent (max.)
d.	Expansion	1 percent (max.)
e.	Expansion inspection sample size	One M24705/1BN60
f.	Fluid immersion	±5 (max.)
g.	Fluid immersion inspection sample size	Three M24705/1BN40
h.	IR signature	Informational
i.	IR signature inspection sample size	One M24705/1BN60

Identification marking. Passage diameter marking shall be required on 20 mm exterior width cable insert blocks and larger. Passage diameter shall not be required on blanking insert blocks. On filler inserts, item designation and manufacturers' name or logo shall be on the inserts; other markings are not required. Optional markings on cable insert blocks shall consist of color code dots for matching insert block halves of a selected module size, cable diameter range in inches, and manufacturers' catalog numbers.

Round frame multiple cable penetrator test specimen (see figure 4). The components for the round multiple cable penetrator test specimen shall be as specified below. Each component listed shall be required for each test specimen. Reconfiguration of cables within the penetrator from that shown on figure 4 requires NAVSEA approval.

a.	Frame	1 – M24705/3F*8 (The * represents frame and hardware material, respectively, which may be CRES or steel (S) in accordance with MIL-DTL-24705.)
b.	End-packing assembly	1 – M24705/3E*8 (The * represents frame and hardware material, respectively, which may be CRES or steel (S) in accordance with MIL-DTL-24705.)
c.	Two-piece square-faced cable insert blocks	3 – M24705/1BN2009/12 1 – M24705/1BN3012/15 4 – M24705/1BN3021/24 2 – M24705/1BN4022/25
d.	Blanking insert blocks	3 – M24705/1BN20 3 – M24705/1BN30
e.	Filler inserts	1 – M24705/1BN40
f.	Cables	2 – LSDSGU-4 2 – LSFSGU-4 2 – LSMSCU-24 3 – LSMSCU-30 1 – LSMDU-6
g.	Lubricant	SAE-AS8660

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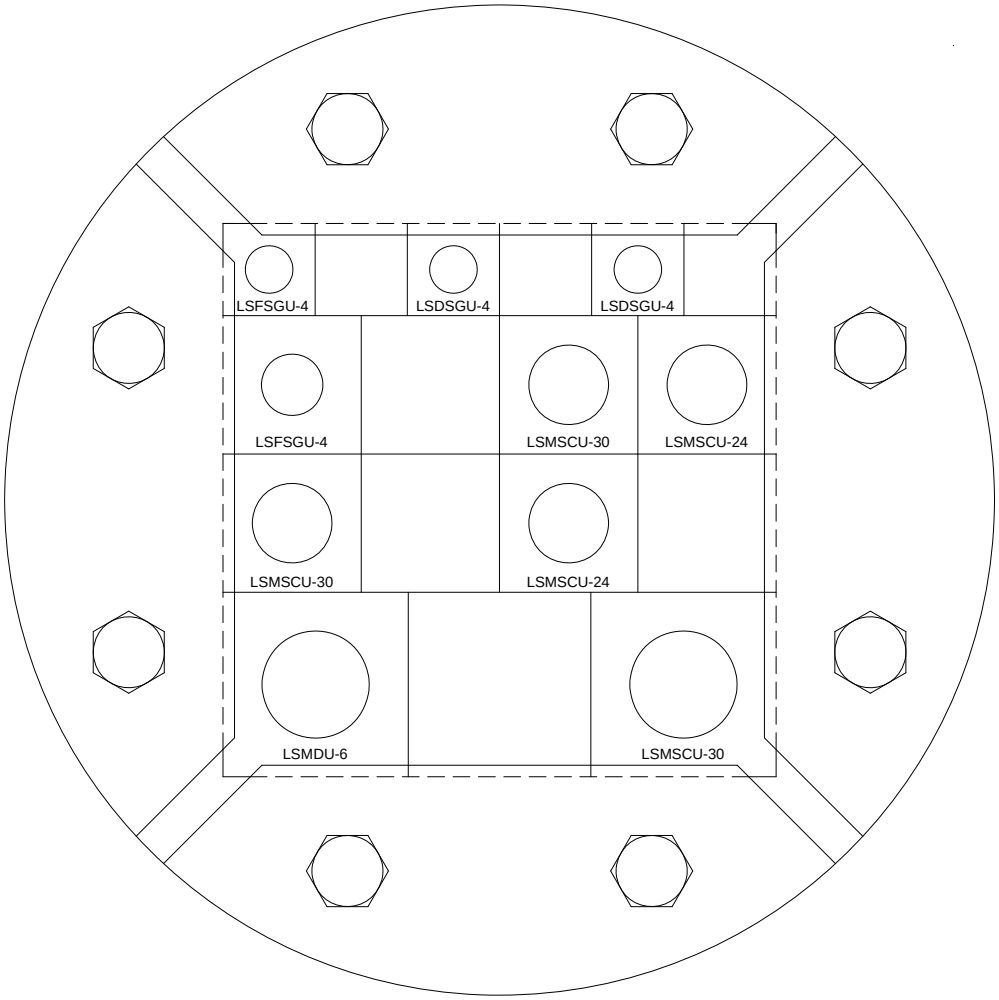


FIGURE 4. Round frame multiple cable penetrator test specimen, cable configuration.

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Navy – SH

Review Activity:
DLA – GS

Preparing Activity:
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(Project 5975-2009-006)

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